

PH.D. QUALIFYING EXAM 2019: PROBABILITY

Rules: This is a four hour exam consisting of two problems, both of which should be answered. You may use ONE page (front and back, 8.5 by 11 inch paper) HANDWRITTEN “cheat sheet”. No other references are allowed. No use of cell phones, calculators, computers, or other electronic devices is permitted for any reason during the time while the exam is being administered, including any time during which you leave the exam room for any reason. **Show your work and justify your answers clearly. If you use known results (e.g., results stated/proven in class), please state them clearly.** Please put your name or initials on every page. Finally, write your answers clearly and legibly on blank papers, preferably using pens, leaving an impression deep enough for the papers to be scanned. At the end of the exam, separate out the bunches of pages for problems 1 and 2, and hand them over to the proctor. Good luck!

Problem 1

Note: The problem has nine parts, **ALL of which should be attempted**. Question 8 asks you to prove an interesting result, an application of which is demonstrated throughout the rest of the problem. Question 9 ties questions 1-7 and question 8 together to drive home the main point behind this entire exercise. **Thus, while deciding upon the order in which you wish to answer the nine parts, you can choose to directly attempt question 8 FIRST without reading and attempting the first seven questions and then return to answering the remaining parts. However, you should choose the order that makes you feel most comfortable.**

Background: For each $N = 1, 2, \dots$, consider a population π_N of N individuals or units for which the values of a variable Y of interest are $Y_{N1}, \dots, Y_{NN}$. For example, the individuals can be human beings and Y can represent their weights or heights. Note that (a) $Y_{N1}, \dots, Y_{NN}$ are fixed quantities and not random variables and (b) two indices are used in the suffix of Y_{Ni} to allow representation of a sequence of populations $\{\pi_N, N = 1, 2, \dots\}$. We are interested in making inference about the population total $\sum_{i=1}^N Y_{Ni}$ using a random sample of size n .

Consider the following two ways of drawing a sample from the population:

- (a) Bernoulli sampling: For $N \geq 1$, each of the units in π_N is included in the sample independently with probability n/N , for some $1 \leq n \leq N$. That is, the i th unit of π_N is included in the sample if and only if $K_i = 1$ for $K_1, K_2, \dots, K_N \stackrel{iid}{\sim} \text{Bern}(n/N)$.
- (b) Simple random sampling: A simple random sample of size n is taken from the population of size N without replacement. This means that all possible ways of drawing n out of N units without replacement are equally likely. Let $J_1, \dots, J_N$ denote the inclusion indicators in this case, i.e., $J_i = 1$ if unit i is included in the sample and $J_i = 0$ otherwise.

Now answer the following questions:

1. (5 points) Write down the joint probability distribution of $K_1, \dots, K_N$ defined in (a) above, that is, $P\{K_1 = k_1, \dots, K_N = k_N\}$. Make sure to explicitly mention for what values of $k_1, \dots, k_N$ this probability will be zero or non-zero.
2. (5 points) Write down the joint probability distribution of $J_1, \dots, J_N$ defined in (b) above, that is, $P\{J_1 = j_1, \dots, J_N = j_N\}$. Make sure to explicitly mention for what values of $j_1, \dots, j_N$ this probability will be zero or non-zero.
3. (6 points) Find $E(J_i)$ and $\text{var}(J_i)$ for fixed $i \in \{1, \dots, N\}$.
4. (4 points) Let $S_N = \sum_{i=1}^N J_i Y_{Ni}$ where $J_1, \dots, J_N$ are defined in (b) above. Write down in words what the quantity S_N represents.

5. (15 points) Define $T_N = \sum_{i=1}^N K_i(Y_{Ni} - \bar{Y}_N) + n\bar{Y}_N$ where $K_1, \dots, K_N$ are defined in (a) above and $\bar{Y}_N = N^{-1} \sum_{i=1}^N Y_{Ni}$ is the population mean. Find the expectation and variance of T_N and show that $E(T_N - S_N) = 0$.

Intermediate information: We will now proceed to calculate the variance of $T_N - S_N$ using an interesting conditioning argument. Suppose we generate the Bernoulli and simple random samples simultaneously. First, we can generate an iid sequence of $\text{Bern}(n/N)$ K_i s to obtain the Bernoulli sample. If $\tilde{n} = \sum_{i=1}^N K_i = n$, we accept this sample. If $\tilde{n} > n$, then we simply choose a simple random subsample of size $\tilde{n} - n$ from this sample, and discard that subsample. If $\tilde{n} < n$, then we choose a simple random sample of size $n - \tilde{n}$ from among the units not selected in the sample, and augment those units to our sample. Then the resulting sample is a simple random sample of size n . Conditioning on the sample size $\tilde{n}$ of the Bernoulli sample, we can argue that the difference between S_N and T_N is a simple random sample of size $\Delta_n = |\tilde{n} - n|$ from the population $\{Y_{N1} - \bar{Y}_N, Y_{N2} - \bar{Y}_N, \dots, Y_{NN} - \bar{Y}_N\}$. Consequently, applying standard results from sample survey, it easily follows that

$$(1) \quad E(T_N - S_N | \tilde{n}) = 0,$$

$$\text{var}(T_N - S_N | \tilde{n}) = \frac{\Delta_n}{N} \frac{N - \Delta_n}{N - 1} \sum_{i=1}^N (Y_{Ni} - \bar{Y}_N)^2,$$

where $\Delta_n = |\tilde{n} - n|$ as defined above.

Now answer parts 6 and 7:

6. (20 points) Using (1), show that

$$\text{var}(T_N - S_N) \leq \sqrt{N \frac{n}{N} \left(1 - \frac{n}{N}\right)} \frac{1}{N} \sum_{i=1}^N (Y_{Ni} - \bar{Y}_N)^2.$$

Hint: Use Eve's law and Jensen's inequality.

7. (10 points) Now show that if both n and $N - n$ tend to infinity as $N \rightarrow \infty$, then

$$\frac{E(T_N - S_N)^2}{\text{var}(T_N)} \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

8. (25 points) In this part we shall prove a famous result called Hájek's lemma. Suppose, we have two sequences of random variables $\{U_n, n = 1, 2, \dots\}$ and $\{V_n, n = 1, 2, \dots\}$, and a random variable L . If

$$\frac{U_n - E(U_n)}{\sqrt{\text{var}(U_n)}} \xrightarrow{d} L \quad \text{and} \quad \frac{E(U_n - V_n)^2}{\text{var}(U_n)} \rightarrow 0,$$

then, $\frac{V_n - E(V_n)}{\sqrt{\text{var}(V_n)}} \xrightarrow{d} L$. Prove this by successively working through the following steps, arguing why they eventually lead to the final result.

(i)

$$\frac{U_n - E(U_n)}{\sqrt{\text{var}(U_n)}} - \frac{V_n - E(V_n)}{\sqrt{\text{var}(U_n)}} \xrightarrow{p} 0.$$

[**Hint:** Show that $E \left(\frac{U_n - E(U_n)}{\sqrt{\text{var}(U_n)}} - \frac{V_n - E(V_n)}{\sqrt{\text{var}(U_n)}} \right)^2 \rightarrow 0$.]

(ii)

$$\frac{V_n - E(V_n)}{\sqrt{\text{var}(U_n)}} \xrightarrow{d} L.$$

(iii)

$$\text{var}(U_n)/\text{var}(V_n) \rightarrow 1.$$

9. (10 points) Now let's get back to the original problem set up. Suppose, under some regularity conditions, and assuming that n and $N - n$ approach infinity as N does, you are able to prove that

$$\frac{T_n - E(T_n)}{\sqrt{\text{var}(T_n)}} \xrightarrow{d} L.$$

Then (i) what can you say about the asymptotic distribution of S_N ?

(ii) In the context of the problem, what does “ n and $N - n$ approach infinity as N does” mean? (iii) Keeping in mind the definition of S_N , what can such a result be called?

Problem 2

Consider the following undirected graph for $n \geq 2$:

- the vertices are $\mathcal{X} = \{-n, \dots, -1, 0, 1, \dots, n\}$,
- the edges are
 - $\{-x, -y\}$ for all positive integers $1 \leq x < y \leq n$ (so that the subgraph made of $\{-n, \dots, -1\}$ is fully connected),
 - $\{x, y\}$ for all positive integers $1 \leq x < y \leq n$ (so that the subgraph made of $\{1, \dots, n\}$ is fully connected),
 - $\{-1, 0\}$ and $\{0, 1\}$ (so that the only path from a negative vertex to a positive vertex is through the vertex 0).

Let P be the matrix of transitions of the simple random walk on this graph and let $(X_t)_{t \geq 0}$ be the corresponding Markov chain. The expectation and probability signs $\mathbf{E}_x[\cdot]$ and $\mathbf{P}_x[\cdot]$ denote probability and expectation when the chain is started from state $x \in \mathcal{X}$.

1. (2.5 points) Draw a picture of the graph for $n = 4$. (*Hint: for verification, the picture should have 9 nodes and 14 edges*).
2. (2.5 points) For general n , how many neighbors do 1 and -1 have? How many neighbors do 2 and -2 have?
3. (15 points) Let $\tau = \min\{t \geq 0 : X_t = 0\}$. By constructing a coupling of Markov Chains that you will clearly describe (both with words and with mathematical symbols), prove that

$$\max_{x \in \mathcal{X}} \|P^t(x, \cdot) - P^t(-x, \cdot)\|_{TV} \leq \mathbf{P}_x(\tau > t).$$

4. *It is advised to solve the next 3 questions on draft paper before writing your answer.*
 - a. (10 points) Derive an upper bound of the form $\max_{x \in \mathcal{X}} \mathbf{P}_x[\tau > 2] \leq 1 - \epsilon_n$ where $\epsilon_n > 0$ is some deterministic constant depending on n only. (You may handle separately the cases $x = \pm 1$ from the rest).
 - b. (10 points) For any $k \geq 0$, derive a geometric upper bound on $\max_{y \in \mathcal{X}} \mathbf{P}_y[\tau > k]$.
 - c. (10 points) Prove an upper bound of the form $\max_{y \in \mathcal{X}} \mathbf{E}_y[\tau] \leq Cn^\alpha$ where α, C are finite absolute constants that do not depend on n . Clearly specify the values of the absolute constants C and α ; it does not matter if $C > 0$ is not optimal, however the exponent α should be the best possible.

5. (10 points) For $x \geq 1$ and $y \geq 1$, clearly prove that both inequalities in

$$\|P^t(x, \cdot) - P^t(y, \cdot)\|_{TV} \leq \|P(x, \cdot) - P(y, \cdot)\|_{TV} \leq \frac{2}{n-1}$$

hold. In the next questions you may use without proof that the same holds if x, y are such that $x \leq -1, y \leq -1$.

6. (15 points) Here and in the next questions, we assume that $n \geq 65$. Show that $\max_{x \neq 0} \max_{y \neq 0} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{TV} \leq 1/16$ for all $t \geq t_0$ for some t_0 that you should express as a function of n .
7. (15 points) By expressing $P^t(0, \cdot)$ as a linear combination of $P^{t-1}(-1, \cdot)$ and $P^{t-1}(1, \cdot)$ or otherwise, show that $\max_{x \in \mathcal{X}} \max_{y \in \mathcal{X}} \|P^t(x, \cdot) - P^t(y, \cdot)\|_{TV} \leq 1/4$ for $t \geq t_1$ for some t_1 that you should express as a function of n .
8. (10 points) (*Open question, if you have time. Your argument does not have to be fully rigorous*) Argue whether or not the order of magnitude of the bound obtained in the previous question is optimal.